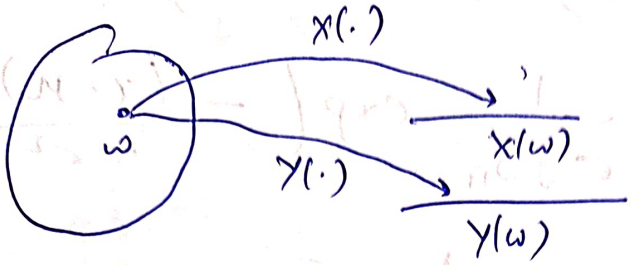
X, Y two RVs on some prob. space $(\Omega, \mathcal{F}, \mathbb{P})$



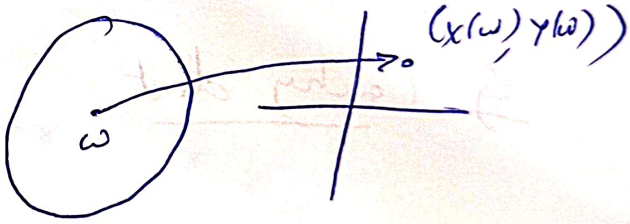
Seen separately, these are two different fns mapping Ω to $\mathbb{R}$

$F_X(\cdot)$, $F_Y(\cdot)$ tell us prob. that X, Y individually take values in some set. But they are fed the same randomness, so the values may be related.

Eg - Student chosen randomly. X = height
Y = weight

Eg - Toss a coin n times. X = # heads, Y = # tails

Think as a random vector

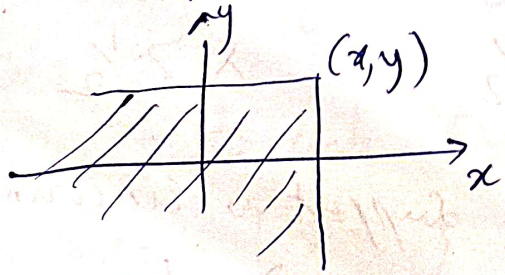


Given a Borel set B in $\mathbb{R}^2$, can we obtain $\mathbb{P}(\{\omega \mid (X(\omega), Y(\omega)) \in B\})$

If X, Y are RV's then $\{\omega \mid (X(\omega), Y(\omega)) \in B\} \in \mathcal{F}$ $\forall B \in \mathcal{B}(\mathbb{R}^2)$

$\mathbb{P}((X,Y) \in B)$ uniquely characterized by joint cdf

$$F_{XY} : \mathbb{R}^2 \to [0,1) \quad . \quad F_{XY}(x,y) = \mathbb{P}(X \le x, Y \le y)$$



This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Properties Joint cdf of (X,Y), $F_{XY}(x,y)$ satisfies

① $\lim_{x,y\to\infty} F_{XY}(x,y) = 1$, $\lim_{\substack{x\to-\infty \\ y\to-\infty}} F_{XY}(x,y) = 0$

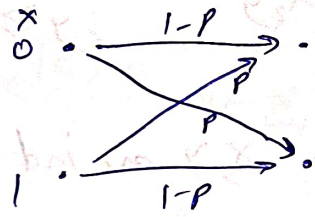
② $x_1 \le x_2,\ y_1 \le y_2 \Rightarrow F_{XY}(x_1,y_1) \le F_{XY}(x_2,y_2)$

③ F_{XY} cont from above $\lim_{y\downarrow 0} F_{XY}(x+u, y+v) = F_{XY}(x,y)$

④ $\lim_{y\to\infty} F_{XY}(x,y) = F_X(x)$
$\lim_{x\to\infty} F_{XY}(x,y) = F_Y(y)$ > marginal CDFs

Joint CDF → Marginal CDFs, not vice versa

Eg → Binary symm channel



$P_X(0) = P_X(1) = 1/2$

$P_Y(0) = P_Y(1) = \frac{1}{2}$

$F_{XY}(0,0) = P_{XY}(0,0) = \frac{1}{2}\cdot(1-p)$

$P_X(x=0)\cdot P_{Y|X}(y=0|x=0)$

Marginal CDF same for any p.
But joint cdf may be very diff.

We can generalize to n-RV's (random vector)

$F_{X_1,X_2\ldots X_n}(x_1,x_2\ldots x_n) = P(X_1 \le x_1,\ X_2 \le x_2 \ldots X_n \le x_n)$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Independence – Random variables $X_1, X_2 \dots X_n$ are independent

if $F_{X_1, X_2 \dots X_n}(x_1, x_2 \dots x_n) = F_{X_1}(x_1) \cdot F_{X_2}(x_2) \dots F_{X_n}(x_n)$

$\forall (x_1, x_2 \dots x_n) \in \mathbb{R}^n$

Technical Result – If X_1, X_2 are ind., then $\forall B_1, B_2 \in \mathcal{B}(\mathbb{R})$

$$P(X_1 \in B_1, X_2 \in B_2) = P(X_1 \in B_1)\, P(X_2 \in B_2)$$

[one side proof easy by choosing $B_1 = (-\infty, x_1]$, $B_2 = (-\infty, x_2]$]

Eg $F_{XY}(x,y) = (1 - e^{-\lambda_1 x} - e^{-\lambda_2 y} + e^{-(\lambda_1 x + \lambda_2 y)})$ $x, y \geq 0$

$\lambda_1, \lambda_2 \geq 0$. Are X, Y ind?

$$F_X(x) = \lim_{y \to \infty} F_{XY}(x,y) = 1 - e^{-\lambda_1 x} \quad x \geq 0$$

$$F_Y(y) = \lim_{x \to \infty} F_{XY}(x,y) = 1 - e^{-\lambda_2 y} \quad y \geq 0$$

$\therefore F_{XY}(x,y) = F_X(x) \cdot F_Y(y) \Rightarrow X, Y$ are ind

---

Discrete Random Vectors

(X, Y) → takes values in some countable set $\Sigma_X \times \Sigma_Y$

described by **joint PMF** $P_{X,Y}(x,y) = P_{XY}(X = x, Y = y)$

① $P_X(x) = \sum_{y \in \Sigma_Y} P_{XY}(x,y)$, ② $P_Y(y) = \sum_{x \in \Sigma_X} P_{XY}(x,y)$

③ $\sum_{(x,y) \in \Sigma_X \times \Sigma_Y} P_{XY}(x,y) = 1$

Joint pmf uniquely specifies joint cdf

Eg

Y \ X	0	1	2
0	0	1/4	1/8
1	1/8	0	1/4
2	1/8	1/8	0

$(i,j)^{th}$ entry → $P(X = i, Y = j)$

$$P(X = x) = \begin{cases} 1/4 & x = 0 \\ 3/8 & x = 1 \\ 3/8 & x = 2 \end{cases}$$

$P(X > Y) = P(1,0) + P(2,0) + P(2,1)$

$= 1/4 + 1/8 + 1/4 = 5/8$

This file is meant for personal use by [illegible]@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Eg - Multinomial dist : collection of k random variables parameterized by $(n, p_1, p_2 \cdots p_k)$

p_i = prob. of an item going to i^{th} bin

$$P(X_1 = n_1, X_2 = n_2 \cdots X_k = n_k) = \frac{n!}{n_1! n_2! \cdots n_k!} p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

$$\sum_{i=1}^{k} n_i = n$$

Conditional PMF

$$P_{X|Y}(x|y) \triangleq P(X = x | Y = y) = \frac{P_{XY}(x,y)}{P_Y(y)} \quad x \in \Sigma_X, \; P_Y(y) > 0$$

① For given y, $P_{X|Y}(\cdot|y)$ is a valid pmf

$$\sum_{x \in \Sigma_X} P_{X|Y}(x|y) = \frac{\sum_{x \in \Sigma_X} P_{XY}(x,y)}{P_Y(y)} = \frac{P_Y(y)}{P_Y(y)} = 1$$

② $$P_X(x) = \sum_{y \in \Sigma_Y} P_{XY}(x,y) = \sum_{y \in \Sigma_Y} P_Y(y) \cdot P_{X|Y}(x|y)$$

Eg One way of visualizing joint PMF is via a table

Y \ X	0	1	2
0	0	1/4	1/8
1	1/8	0	1/4
2	1/8	1/8	0

$$P(X > Y) = \frac{1}{4} + \frac{1}{8} + \frac{1}{4} = \frac{5}{8}$$

Prob of an event is the sum of pmf values over respective region

$$P_{X|Y}(0|2) = \frac{P_{XY}(x=0, y=2)}{P_Y(y=2)} = \frac{1/8}{\frac{1}{8} + \frac{1}{8} + 0} = \frac{1}{2}$$

$$P_{X|Y}(1|2) = 1/2, \quad P_{X|Y}(2|2) = 0$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

These 3 conds are equivalent

① X, Y are ind. ② $P_{XY}(x,y) = P_X(x) \cdot P_Y(y)$

③ $P_{X|Y}(x|y) = P_X(x)$ when $P_Y(y) > 0$

Proof $2 \Leftrightarrow 3$ from defn $\quad 2 \to 3$ ∎

$1 \Rightarrow 2$: $P(X \in B_1, Y \in B_2) = P(X \in B_1) \cdot P(Y \in B_2)$

Take $B_1 = \{x\}$, $B_2 = \{y\}$

$2 \Rightarrow 1$ $\quad F_{XY}(x,y) = \sum_{\substack{u \in \Sigma_X, u \le x \\ u \in \Sigma_Y, v \le y}} \underbrace{P_{XY}(u,v)}_{P_X(u) \cdot P_Y(v)}$

$$= \left(\sum_{\substack{u \le x \\ u \in \Sigma_X}} P_X(u) \right) \left(\sum_{\substack{v \le y \\ v \in \Sigma_Y}} P_Y(v) \right)$$

$$= F_X(x) \cdot F_Y(y)$$

Eg Coin, $P(\text{head}) = p$. Toss coin n times

X = # heads, Y = # tails

$$P(X=x, Y=y) = \begin{cases} {}^nC_x \, p^x (1-p)^y & \text{if } x+y = n,\ x, y \ge 0 \\ 0 & \text{o.w} \end{cases}$$

Are X, Y independent? Find $P_X(x)$, $P_Y(y)$

No. Compare to $P(X=x, Y=y)$

Now take $N \sim \text{Poi}(\lambda)$

$$P_{XY}(m,n) = P(X=m, Y=n) = P(X=m, Y=n, N=m+n)$$

$$= P(N=m+n)\, P(X=m, Y=n \mid N=m+n)$$

$$= \frac{e^{-\lambda} \lambda^{m+n}}{(m+n)!} \binom{m+n}{n} p^m (1-p)^n = \frac{e^{-\lambda p} e^{-\lambda(1-p)} (\lambda p)^m (\lambda(1-p))^n}{n!\, m!}$$

$$= \frac{e^{-\lambda p} (\lambda p)^m}{m!} \cdot \frac{e^{-\lambda(1-p)} (\lambda(1-p))^n}{n!} = P(X=m) \cdot P(Y=n)$$

why? why?

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Jointly Continuous RVs

X, Y are said to be jointly cts if $\exists$ non-neg. $f_{XY}: \mathbb{R}^2 \to [0,\infty)$ s.t. $F_{XY}(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{XY}(s,t)\,ds\,dt$

F_{XY}: joint cdf

f_{XY}: joint pdf

Note: $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(s,t)\,ds\,dt = 1$

X, Y jointly cts $\quad F_X(x) = \lim_{y\to\infty} F_{XY}(x,y)$

$= \lim_{y\to\infty} \int_{-\infty}^{x}\int_{-\infty}^{y} f_{XY}(s,t)\,ds\,dt = \int_{-\infty}^{x}\int_{-\infty}^{\infty} f_{XY}(s,t)\,ds\,dt$

$= \int_{-\infty}^{x} f_X(s)\,ds$

So able to write $F_X(x) = \int_{-\infty}^{x} f_X(s)\,ds$

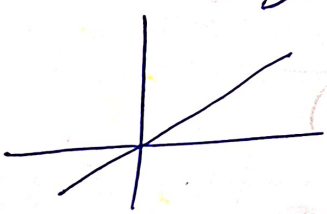
So X & Y are marginally cts too.

$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,t)\,dt$

$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(s,y)\,ds$

$P((X,Y)\in B) = 0$ if B is a Borel set in $\mathbb{R}^2$ with zero Lebesgue measure

Converse not necessarily true. X, Y could be marginally cts but not jointly cts. Eg $X \sim N(0,1)$, $Y = 2X$



(X,Y) takes values on the line $\{(x,y)\in\mathbb{R}^2 / y = 2x\}$

If joint density existed, then $\int_{L} f_{XY}(s,t)\,ds\,dt = 1$

but L has zero Lebesgue measure in $\mathbb{R}^2$. So should have measure 0

Contradiction.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Eg - Jointly Gaussian RV

Eg Do Exp RV's, also. Next page bottom.

(X, Y) - standard bivariate Gaussian RV if

$$f_{XY}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\left[\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right]\right)$$

for $\rho \in [-1, 1]$

X, Y marginally $\sim N(0,1)$ standard Gaussians

Independence

X, Y jointly cts.

X, Y ind. if $F_{XY}(x,y) = F_X(x) F_Y(y) \quad \forall x, y \in \mathbb{R}$

$$\iff \int_{-\infty}^{x}\int_{-\infty}^{y} f_{XY}(s,t)\,ds\,dt = \left(\int_{-\infty}^{x} f_X(s)\,ds\right)\left(\int_{-\infty}^{y} f_Y(t)\,dt\right)$$

$$= \int_{-\infty}^{x}\int_{-\infty}^{y} f_X(s) f_Y(t)\,ds\,dt$$

True for $\forall x, y$ iff $\boxed{f_{X,Y}(s,t) = f_X(s) f_Y(t)} \quad \forall s, t$

Conditional CDF & PDF

Want to define $F_{X|Y}(x|y) = \mathbb{P}(X \le x \mid Y = y)$. But $\mathbb{P}(Y=y) = 0$ since Y is cts. Instead we condition on the event $\mathbb{P}(Y \in [y, y+\varepsilon])$ for small ε; note that $\mathbb{P}(Y \in [y, y+\varepsilon]) = \int_y^{y+\varepsilon} f_Y(t)\,dt \approx f_Y(y)\cdot\varepsilon$

Note $\mathbb{P}(X \le x \mid Y \in [y, y+\varepsilon]) = \dfrac{\mathbb{P}(X \le x, Y \in [y, y+\varepsilon])}{\mathbb{P}(Y \in [y, y+\varepsilon])}$

$$= \frac{\int_{-\infty}^{x}\left(\int_y^{y+\varepsilon} f_{X,Y}(s,t)\,dt\right)da}{\int_y^{y+\varepsilon} f_Y(t)\,dt} \approx \frac{\int_{-\infty}^{x} f_{X,Y}(s,y)\,da}{f_Y(y)}$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

This motivates the following defn

Defn - Given jointly cont RVs (X,Y). The conditional CDF of X given Y is defined as.

$$F_{X|Y}(x|y) = \frac{\int_{-\infty}^{x} f_{X,Y}(s,y)\,ds}{f_Y(y)}, \text{ when } f_Y(y) > 0$$

Conditional PDF defined as $f_{X|Y}(x|y) \triangleq \frac{f_{X,Y}(x,y)}{f_Y(y)}$ when $f_Y(y) > 0$

① $f_{X|Y}(x,y)$ describes density of X given Y close to y.

② Note similarity with conditional pmf which was $P_{X|Y}(x|y) \triangleq \frac{P_{X,Y}(x,y)}{P_Y(y)}$

③ $\int_{-\infty}^{\infty} f_{X|Y}(x|y)\,dx = 1$

$$f_X(x) = \int_{-\infty}^{\infty} f_Y(y)\, f_{X|Y}(x|y)\,dy$$

Eg Jointly cont RVs. $F_{X,Y}(x,y) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-(x+y)}, & x,y \geq 0 \\ 0 & \text{o.w.} \end{cases}$

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$$

$$\frac{\partial}{\partial y} F_{X,Y}(x,y) = e^{-y} - e^{-(x+y)}$$

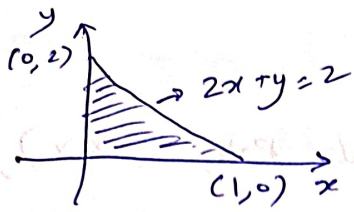
$$\frac{\partial}{\partial x} = e^{-(x+y)}, \quad x,y \geq 0$$

So $f_{X,Y}(x,y) = e^{-(x+y)}$ $x,y \geq 0$

Marginal pdfs. : $f_X(x) = e^{-x}$, $x \geq 0$

$f_Y(y) = e^{-y}$ $y \geq 0$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

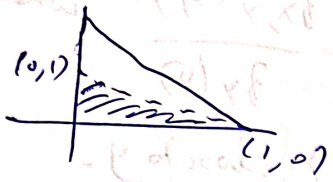
Eg 

$$f_{X,Y}(x,y) = \begin{cases} 1 & \text{in shaded rg.} \\ 0 & 0.w. \end{cases}$$

Find ① $P(X+Y \le 1)$, ② $f_X(\cdot)$, $f_Y(\cdot)$,

③ $f_{X|Y}(\cdot|y)$, $f_{Y|X}(\cdot|x)$

① $P(X+Y \le 1) = \int_{x=0}^{1} \int_{y=0}^{1-x} 1 \, dy \, dx$



$$= \int_{x=0}^{1} (1-x)\,dx = \int_0^1 z\,dz = \frac{1}{2}$$

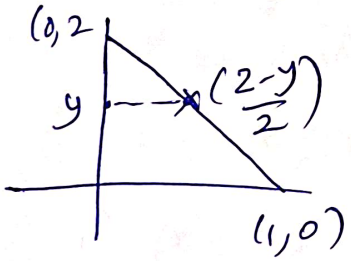
② Marginals. $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,t)\,dt$

$$= \int_0^{2-2x} 1 \cdot dt = 2-2x, \quad x \in [0,1]$$

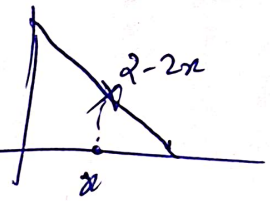
$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(z,y)\,dz = \int_0^{(2-y)/2} 1 \cdot dz$$

$$= \frac{2-y}{2}, \quad 0 \le y \le 2$$

③ Conditional pdf $f_{X|Y}(x|y) = \dfrac{f_{X,Y}(x,y)}{f_Y(y)} =$



$$= \begin{cases} \dfrac{2}{2-y} & 0 \le x \le \dfrac{2-y}{2} \\ 0 & 0.w. \end{cases}$$



$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

$$= \begin{cases} \dfrac{1}{2-2x} & 0 \le y \le 2-2x \\ 0 & 0.w. \end{cases}$$

Conditional density is uniform

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Eg $X \sim \lambda_1 e^{-\lambda_1 x}, \; x \geq 0 \qquad Y \sim \lambda_2 e^{-\lambda_2 y}, \; y \geq 0$

X, Y independent

$$P(X \leq Y) = \int_{y=0}^{\infty} \int_{x=0}^{y} \lambda_1 \lambda_2 e^{-\lambda_1 x} e^{-\lambda_2 y} \, dx \, dy$$

$$= \int_{y=0}^{\infty} \lambda_2 e^{-\lambda_2 y} \left(\int_{x=0}^{y} \lambda_1 e^{-\lambda_1 x} \, dx \right) dy$$

$$= \int_{y=0}^{\infty} \lambda_2 e^{-\lambda_2 y} \left(1 - e^{-\lambda_1 y}\right) dy$$

$$= 1 - \frac{\lambda_2}{\lambda_1 + \lambda_2} \int_{y=0}^{\infty} (\lambda_1 + \lambda_2) e^{-(\lambda_1 + \lambda_2) y} \, dy$$

$$= \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.